

Question:

In a dance class of 20 students, 20% enrolled in contemporary dance, 25% of the remaining enrolled in jazz dance, and the rest enrolled in hip-hop dance. What percentage of the entire students enrolled in hip-hop dance?

Reasoning + Answer:

<think> To solve this problem , let 's first calculate the number of students enrolled in contemporary dance . Given that there are 2 0 students in total and 2 0 % of them enrolled in contemporary dance , we can find this number by calculating (20×0.20) . So , let's compute $(20 \times 0.20 = 4)$. This means 4 students enrolled in contemporary dance .

Next , we need to determine the number of students remaining after those enrolled in contemporary dance have been subtracted from the total . So , we subtract the 4 students from the total of 2 0 students : $(20 - 4 = 16)$. This leaves us with 1 6 students who are left to be enrolled in either jazz dance or hip-hop dance .

Now , we focus on the jazz dance enrollment . The problem states that 2 5 % of the remaining students enrolled in jazz dance . Since we know there are 1 6 students left after contemporary dance enrollment , we calculate (16×0.25) to find out how many students enrolled in jazz dance . So , let's compute $(16 \times 0.25 = 4)$. This means 4 students enrolled in jazz dance .

Finally , to find out how many students enrolled in hip-hop dance , we need to subtract the number of students enrolled in contemporary dance and jazz dance from the total number of students . We subtract the 4 students from contemporary dance and the 4 students from jazz dance from the remaining 1 6 students : $(16 - 4 - 4 = 8)$. This tells us that 8 students enrolled in hip-hop dance .

Now , to find the percentage of the entire class that enrolled in hip-hop dance , we use the formula for percentage : $(\frac{\text{Number of students in hip-hop dance}}{\text{Total number of students}} \times 100)$. Sub stit uting the numbers we have , we get $(\frac{8}{20} \times 100)$. Calculating this , $(\frac{8}{20} \times 100 = 40)$. </think>

<answer> 4 0 </answer>

